SoyMate: phone-photo quality control for pulses

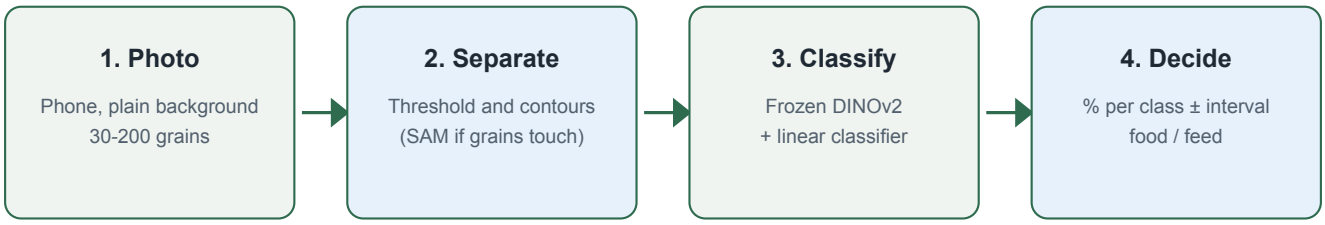
Team plan for Hack the Fork Paris 2026 (Track 3). Short version: the full context, with a source for every figure, is in CONTEXT.md.

In one line: the intake technician at a grain collector takes a photo of a sample of peas or lentils, and SoyMate separates every grain, classifies it (sound, broken, holed, stained, foreign material) and says whether the lot goes to human food or animal feed.

Why

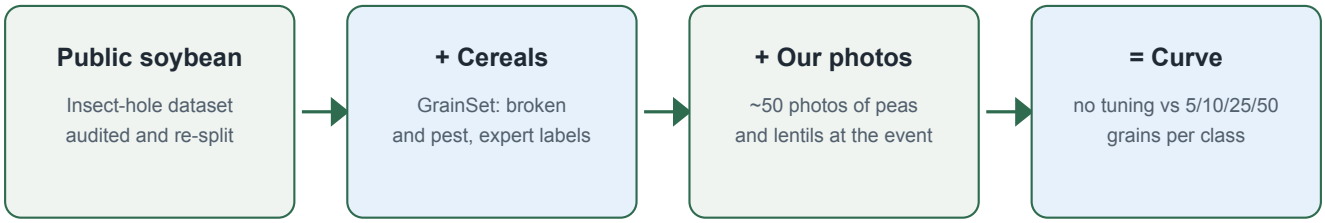
- In the 2025 national quality survey, lentil lots had **on average 24.4 % bruchid-damaged grains**, the highest value since the survey began. Since July 2024 no insecticide is authorised in the field against bruchids in dry pulses.
- In pea, only **55 % of lots** had less than 1 % broken grains, the most limiting criterion for human food.
- French contracts for human food allowed at most **1 % bruchid-damaged grains** (Incograin, 2003 text; whether it is still current must be confirmed). Today this is decided by looking at and counting grains by hand.
- Inarix (France) already does this by phone for cereals; **we found no one doing it for pulses**.

The flow



Everything runs on CPU (the event's AWS account has no GPU). The pretrained model extracts features for each grain and we only train a small classifier: fast, and it works with little data.

How we train without pulse data



We found no public image dataset of damaged pulses. We train on soybean and cereals and **measure** how much transfers to peas and lentils. That curve is the core of the pitch: it turns "with a good dataset we can extend it" into a number.

Data

Dataset	What it gives	Watch out
Soybean pulse beetle (Mendeley)	2,726 photos, insect exit holes	Use our leak-free split (the published one leaks 80 %); crop each grain (the background gives the damage away)
GrainSet (cereals)	Broken, pest-damaged and foreign material, expert labels	Split by capture batch
corn1000 (cereals)	Phone photos with hundreds of grains and exact counts	Non-commercial licence

Radio_lentille (Terres Inovia)	X-rays of bruchid-damaged French lentils	Request access on Hugging Face today
Our photos	Real peas and lentils: sound, broken, (holed)	One class per photo; split by photo

What we must have

Must have

- Classifier trained on soybean, evaluated on the leak-free test set
- ~50 own photos of peas and lentils (at least sound and broken)
- Transfer curve soybean → pulses
- Grain separation in multi-grain photos
- Lot-level % with interval and threshold decision
- Working demo: upload photo → result
- Data provenance table and list of pre-existing resources

Nice to have

+ Real holes in lentil or pea (grains from INRAE)
+ SAM or Cellpose for touching grains
+ Validation with Terres Inovia X-rays
+ Phone demo with a public URL
+ Faba bean and chickpea

Roles

Role	Owns
A: data and evaluation	Soybean data with the leak-free split, photo session, labelling, metrics and provenance table
B: model	Grain separation, features, classifier, transfer curve, lot-level aggregation
C: product and pitch	Demo, mentor questions, business context, deck, backup video and rehearsal

Schedule

When	A (data)	B (model)	C (product)
Fri 20:45-23:00	Download soybean data, apply the leak-free split, crop grains	Download GrainSet; test grain separation on a couple of photos	Mentor questions; declare the track; set up the photo station
Sat 10:15-13:00	Photo session (~50) and labelling	Soybean classifier; leak-free test; transfer without tuning	Demo skeleton; ask for real damaged grains
Sat 13:00	**Checkpoint 1:** the classifier works on soybean and grain separation works		
Sat 14:00-18:00	Mixed photos counted by hand	Curve 5/10/25/50; lot-level %; decision	Demo wired to the model; problem and business slides
Sat 16:00 / 18:00	**Checkpoint 2:** curve ready. **Checkpoint 3:** demo works end to end		
Sat night	Final metrics and figures	Robustness; freeze the model	Deck, backup video, 4-min rehearsal
Sun 8:30-9:30	**Submit before 9:30**: pitch PDF + online project sheet		
Sun 12:00	**Back up everything outside AWS** (GitHub + USB)		

If time runs short, cut in this order: faba bean and chickpea → X-rays → SAM → phone demo → stained class. Never cut: the leak-free split, the curve or the demo.

Photo protocol (~2 h)

- Matte blue or black card, phone fixed at 25-30 cm, diffuse light without flash, a 1 € coin for scale.
- 30-80 grains per photo, not touching; for the demo, one photo with ~200 grains.
- One class per photo (the label comes from the folder): ~2 species × 6 classes × 4 photos ≈ 50 photos.
- Label with Label Studio or CVAT (free Roboflow makes the dataset public).
- Broken grains: we break them ourselves (real damage). Grains pierced by hand: separate class, declared as synthetic.

The 4-minute pitch

Time	What we say
0:00-0:40	**Problem:** in 2025, lentil lots had on average 24.4 % bruchid-damaged grains, with no field insecticide left; for human food the contract allowed at most 1 %; today it is counted by hand
0:40-1:20	**Solution:** one photo; SoyMate separates, classifies and decides
1:20-2:30	**Live demo** with peas and lentils we brought
2:30-3:10	**How:** no pulse data exists; audited soybean data (80 % leakage fixed) + transfer curve with our own photos
3:10-4:00	**Scale:** 477 grain collectors; Inarix shows they pay for this in cereals and we found no one in pulses; the app improves with every intake. Limit: we see the hole, not the larva

Key: the product and the demo are about **French peas and lentils**. Soybean appears only on the "how we built it" slide. If the focus were soybean, we would be close to the animal-feed exclusion and competing head-on with Inarix.

To do before tonight

[ ] Request access to **Radio_lentille** on Hugging Face (JeanEudesH/Radio_lentille) and download it.

[ ] Buy green and coral lentils, split and whole peas, faba beans, chickpeas, soybeans; blue or black card; a phone stand.

All figures and their sources are in CONTEXT.md (§17 Sources, §18 Claim register).

To confirm on Friday before 23:00

[ ] With the organisers: that "Insects" (excluded topics) does not cover detecting insect damage in grains.

[ ] With the organisers: that training on soybean for human food is in scope, and that buying supplies before the event is allowed.

[ ] With mentors: how intake works today (per truck or composite sample, how long it takes) and what tolerances contracts set.

[ ] Ask **Antoine Gardarin** (UMR Agronomie, INRAE, same Palaiseau campus) for real bruchid-damaged grains.